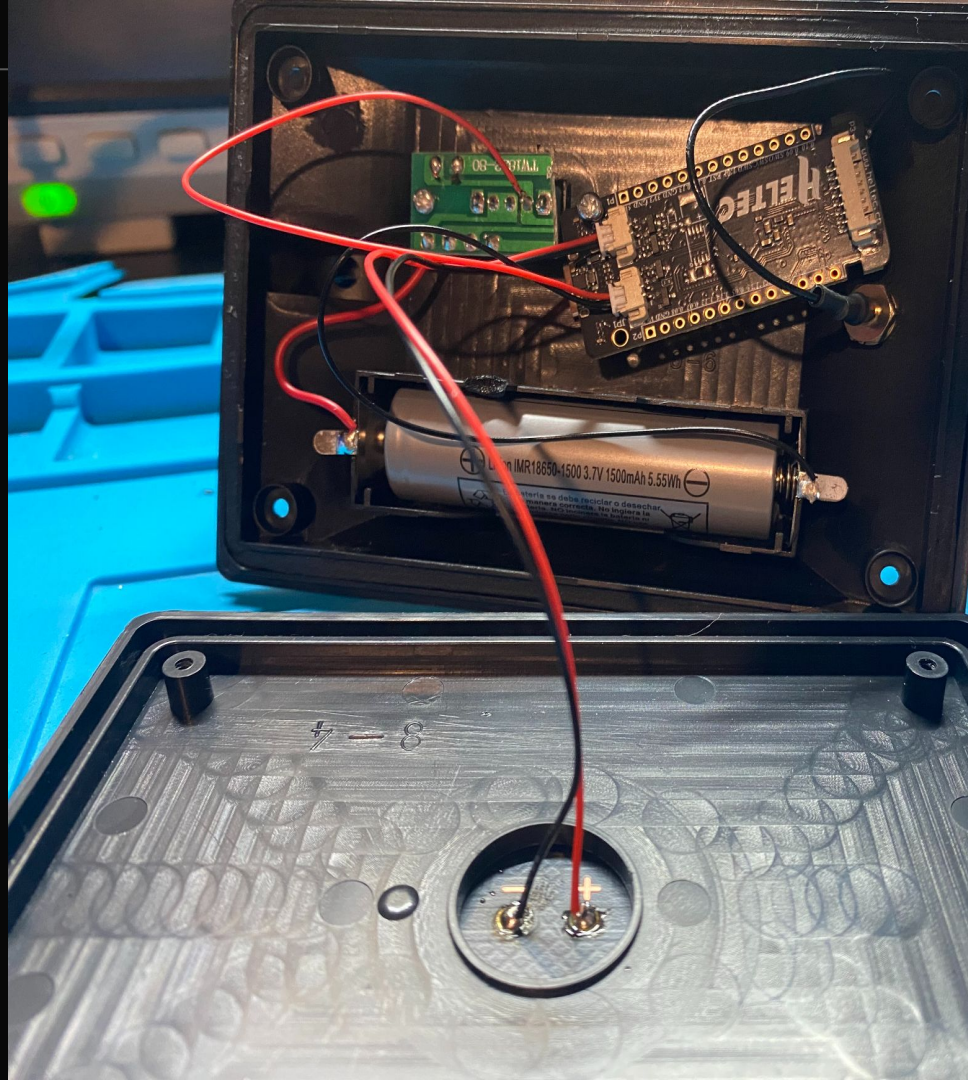
› prepared by Danielle McGuire
for BSidesPGH 2026

mesh hack tricks

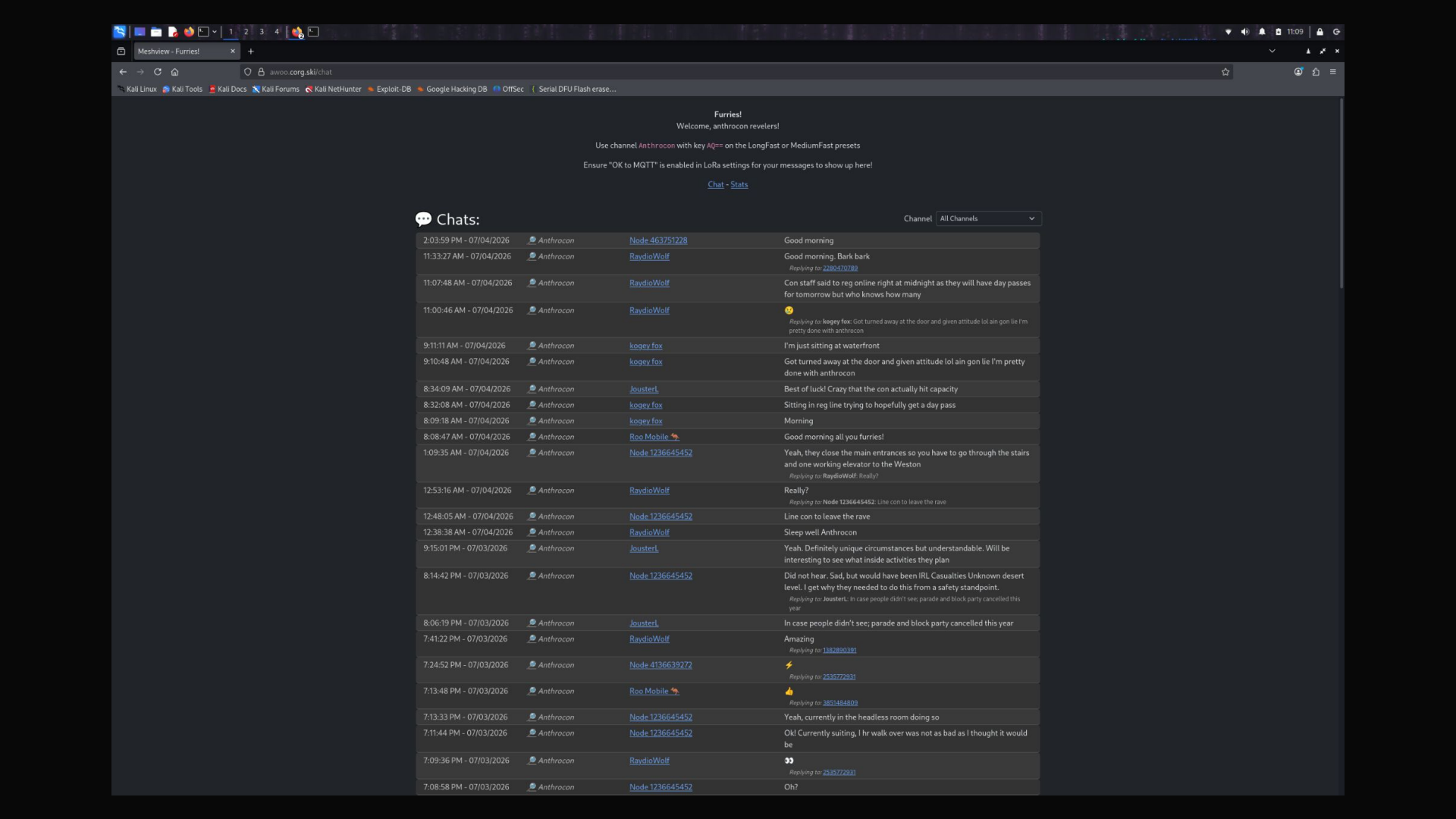


Thank you to the Protomesh Collective and our collaborators!!

Alpine Creature
Brynn Flowers
Creative Chemical
Emi
Giselle J
Green Dragon House
Greyfox
Jamie Werger
Katelyn Chirdon
Milkweed
Rushi Null
Prototype Makerspace
Sunbear

agenda

Intro
What is Meshtastic?
DEMO - Sniffing Wireless Meshtastic Traffic with an RTL-SDR
Radio 101
Why Meshtastic?
Why not Meshtastic?
DEMO – Spoofing Hardware IDs to Exploit Lack of Authentication in Meshtastic Protocol
Building Meshtastic nodes for cheap
DEMO – Programmatic Control via Meshtastic SDK
Monitoring Meshtastic networks
Conclusion



Meshview - Furries!

awoo.corg.ski/chat

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Serial DFU Flash erase...

Furries!
Welcome, anthrocon revelers!

Use channel `Anthrocon` with key `AQ==` on the LongFast or MediumFast presets

Ensure "OK to MQTT" is enabled in LoRa settings for your messages to show up here!

Chat - Stats

💬 Chats:

Channel: All Channels

2:03:59 PM - 07/04/2026	Anthrocon	Node 463751228	Good morning
11:33:27 AM - 07/04/2026	Anthrocon	RaydioWolf	Good morning. Bark bark *Replying to: 2280470789*
11:07:48 AM - 07/04/2026	Anthrocon	RaydioWolf	Con staff said to reg online right at midnight as they will have day passes for tomorrow but who knows how many
11:00:46 AM - 07/04/2026	Anthrocon	RaydioWolf	😢 *Replying to: kogey fox: Got turned away at the door and given attitude lol ain gon lie I'm pretty done with anthrocon*
9:11:11 AM - 07/04/2026	Anthrocon	kogey fox	I'm just sitting at waterfront
9:10:48 AM - 07/04/2026	Anthrocon	kogey fox	Got turned away at the door and given attitude lol ain gon lie I'm pretty done with anthrocon
8:34:09 AM - 07/04/2026	Anthrocon	JousterL	Best of luck! Crazy that the con actually hit capacity
8:32:08 AM - 07/04/2026	Anthrocon	kogey fox	Sitting in reg line trying to hopefully get a day pass
8:09:18 AM - 07/04/2026	Anthrocon	kogey fox	Morning
8:08:47 AM - 07/04/2026	Anthrocon	Roo Mobile 🦘	Good morning all you furries!
1:09:35 AM - 07/04/2026	Anthrocon	Node 1236645452	Yeah, they close the main entrances so you have to go through the stairs and one working elevator to the Weston *Replying to: RaydioWolf: Really?*
12:53:16 AM - 07/04/2026	Anthrocon	RaydioWolf	Really? *Replying to: Node 1236645452: Line con to leave the rave*
12:48:05 AM - 07/04/2026	Anthrocon	Node 1236645452	Line con to leave the rave
12:38:38 AM - 07/04/2026	Anthrocon	RaydioWolf	Sleep well Anthrocon
9:15:01 PM - 07/03/2026	Anthrocon	JousterL	Yeah. Definitely unique circumstances but understandable. Will be interesting to see what inside activities they plan
8:14:42 PM - 07/03/2026	Anthrocon	Node 1236645452	Did not hear. Sad, but would have been IRL Casualties Unknown desert level. I get why they needed to do this from a safety standpoint. *Replying to: JousterL: In case people didn't see; parade and block party cancelled this year*
8:06:19 PM - 07/03/2026	Anthrocon	JousterL	In case people didn't see; parade and block party cancelled this year
7:41:22 PM - 07/03/2026	Anthrocon	RaydioWolf	Amazing *Replying to: 1382890391*
7:24:52 PM - 07/03/2026	Anthrocon	Node 4136639272	⚡ *Replying to: 2535772831*
7:13:48 PM - 07/03/2026	Anthrocon	Roo Mobile 🦘	👍 *Replying to: 3851484809*
7:13:33 PM - 07/03/2026	Anthrocon	Node 1236645452	Yeah, currently in the headless room doing so
7:11:44 PM - 07/03/2026	Anthrocon	Node 1236645452	Ok! Currently suiting, I hr walk over was not as bad as I thought it would be
7:09:36 PM - 07/03/2026	Anthrocon	RaydioWolf	👀 *Replying to: 2535772831*
7:08:58 PM - 07/03/2026	Anthrocon	Node 1236645452	Oh?

i’m about to try and make you as excited about radio as i am.

i'm about to try and make you as excited about radio as i am.

i'm going to tell you about all the amazing and beautiful things happening on the 915MHz (33cm) band

i'm about to try and make you as excited about radio as i am.

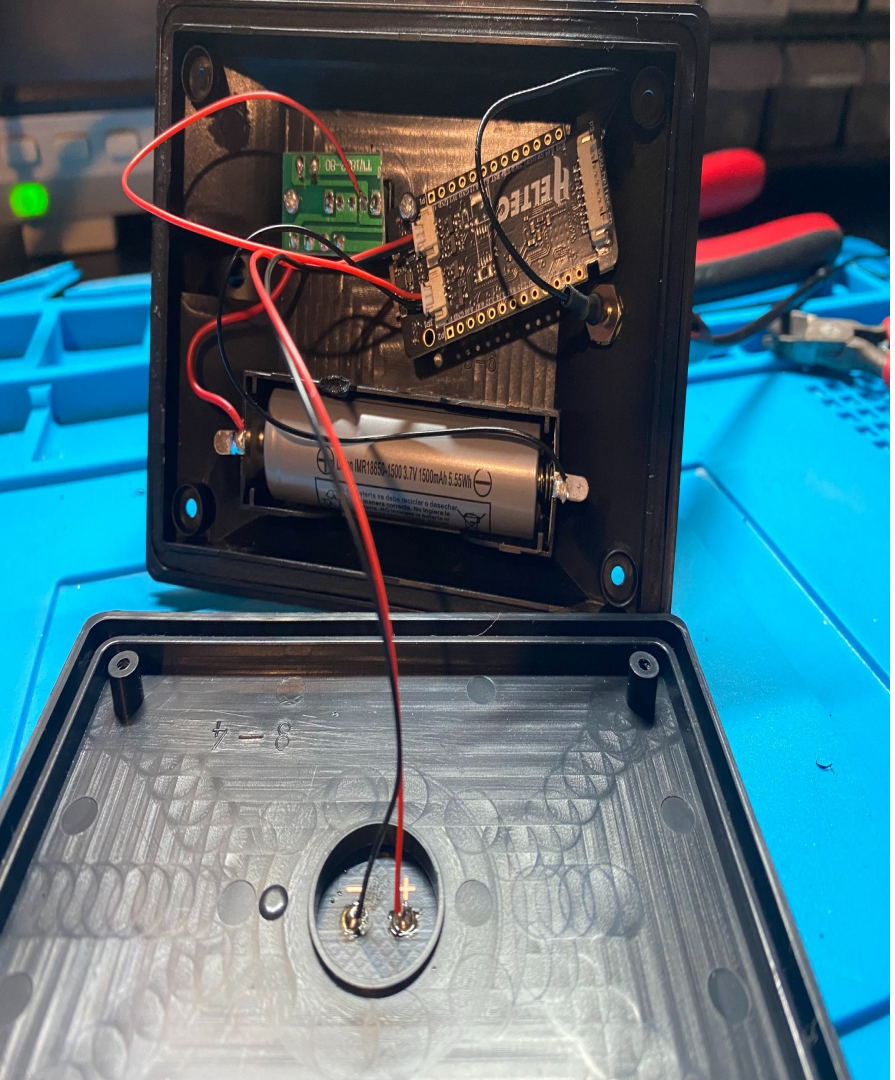
i'm going to tell you about all the amazing and beautiful things happening on the 915MHz (33cm) band and how you can be a part of them.

what is meshtastic?

An open source, off-grid, decentralized mesh network built to run on affordable, low-power devices. ***No cell towers. No internet.*** *Just pure peer-to-peer connectivity.*
front page of https://meshtastic.org

an encrypted communications network that can reliably function without the **Internet**, cellular infrastructure, satellites, or even the power grid - when all else fails, there's the 915





under the hood

Physical layer - wireless **LoRa-modulated** radio waves over suitable unregulated frequency band (915MHz/33cm is designated for 'Industrial, Scientific or Medical' usage by the US FCC)

Application Layer - lots of choices!

- LoRaWan - Internet WAN for IoT
- **Meshtastic** – resilient encrypted decentralized text messaging (simple)
- Meshcore – resilient encrypted decentralized text messaging (powerful)
- Reticulum - Crypto-first physical-medium-agnostic decentralized Internet of the future

MESHTASTIC

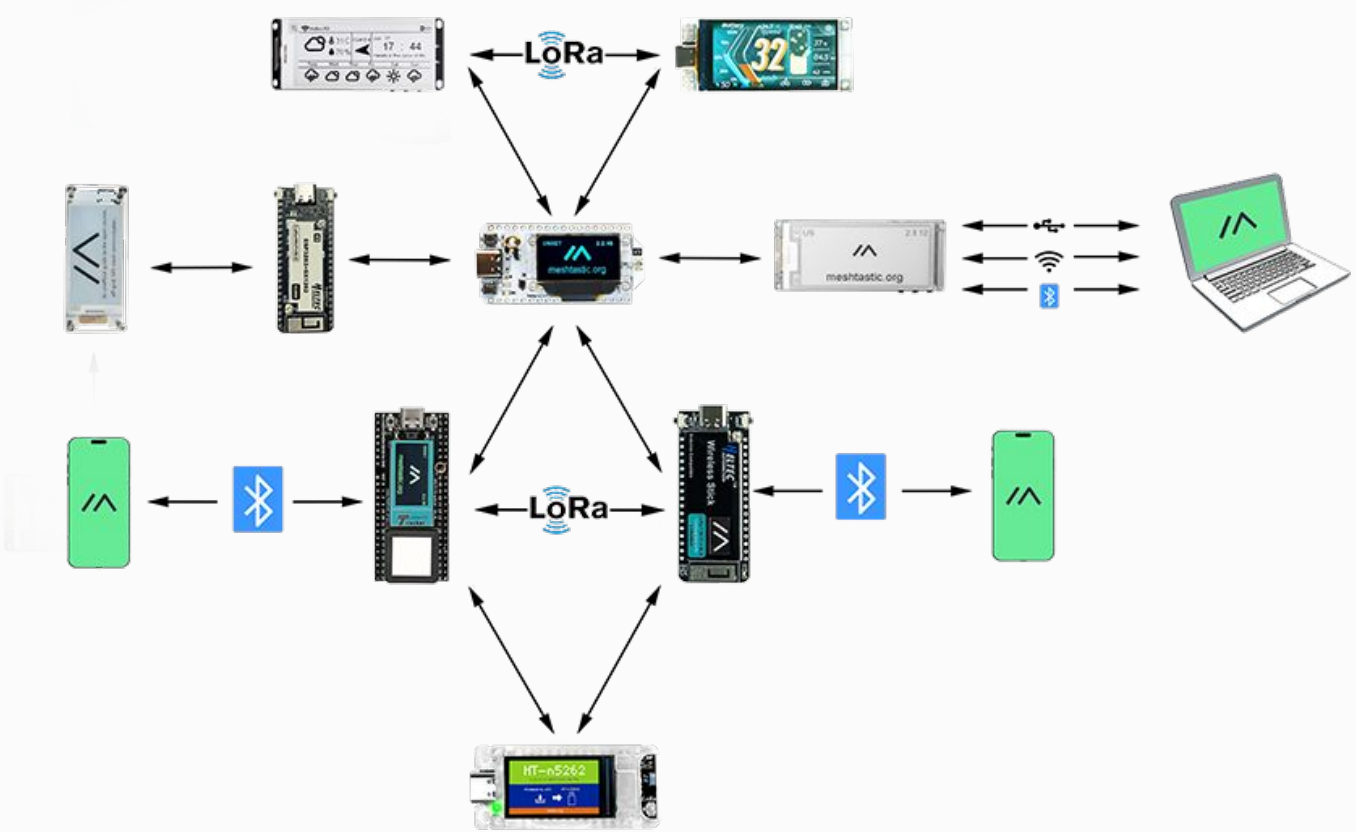
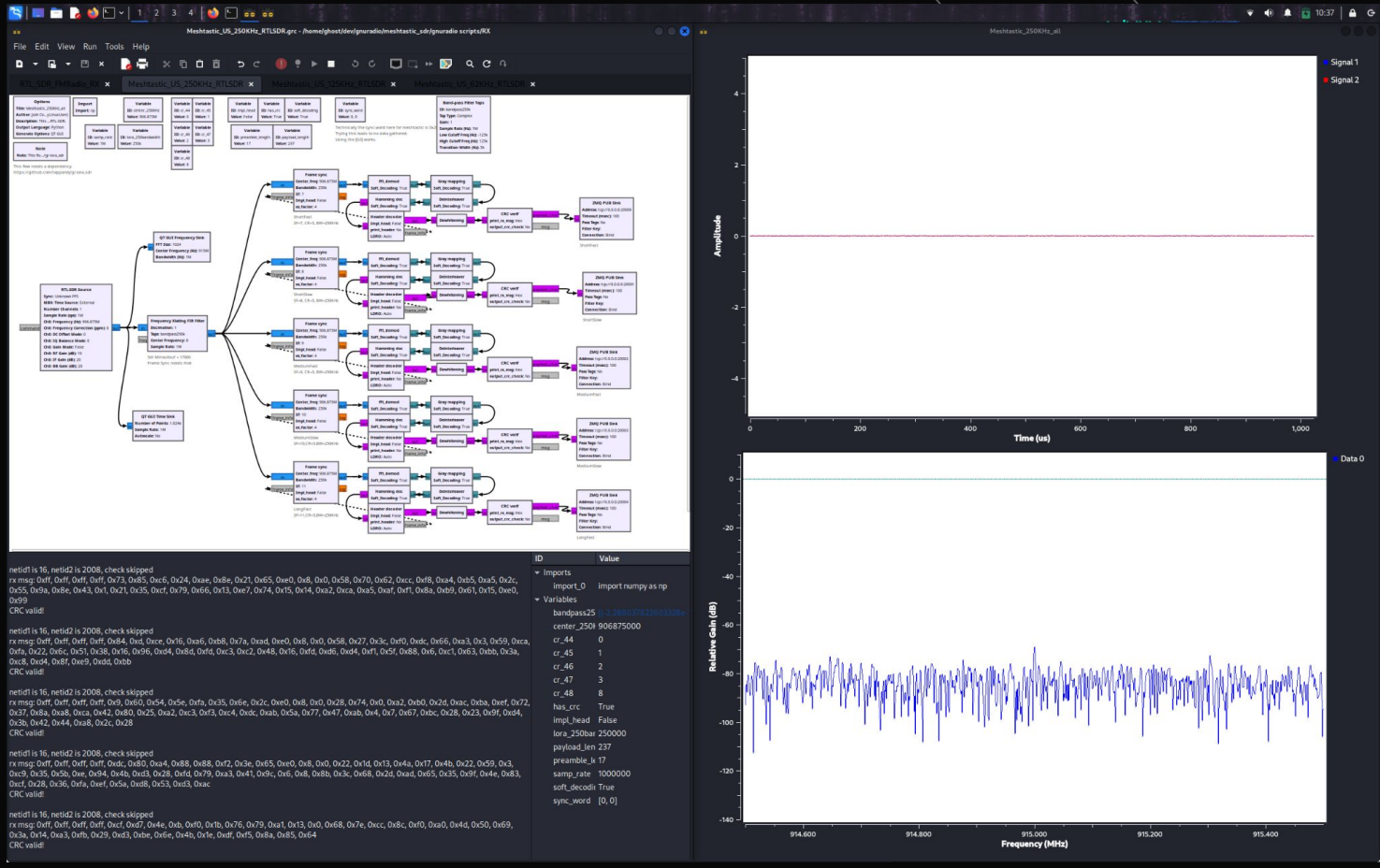


diagram taken from https://heltec.org/eadge-computing/

DEMO - Wirelessly Sniffing Meshtastic Traffic with an RTL-SDR




```
netid1 is 16, netid2 is 2008, check skipped
rx msg: 0xff, 0xff, 0xff, 0xff, 0x73, 0x85, 0xc6, 0x24, 0xae, 0x8e, 0x21, 0x65, 0xe0, 0x8, 0x0, 0x58, 0x70, 0x62, 0xcc, 0xf8, 0xa4, 0xb5, 0xa5, 0x2c, 0x55, 0x9a, 0x8e, 0x43, 0x1, 0x21, 0x35, 0xcf, 0x79, 0x66, 0x13, 0xe7, 0x74, 0x15, 0x14, 0xa2, 0xca, 0xa5, 0xaf, 0xf1, 0x8a, 0xb9, 0x61, 0x15, 0xe0, 0x99
CRC valid!

netid1 is 16, netid2 is 2008, check skipped
rx msg: 0xff, 0xff, 0xff, 0xff, 0x84, 0xd, 0xce, 0x16, 0xa6, 0xb8, 0x7a, 0xad, 0xe0, 0x8, 0x0, 0x58, 0x27, 0x3c, 0xf0, 0xdc, 0x66, 0xa3, 0x3, 0x59, 0xca, 0xfa, 0x22, 0x6c, 0x51, 0x38, 0x16, 0x96, 0xd4, 0x8d, 0xfd, 0xc3, 0xc2, 0x48, 0x16, 0xfd, 0xd6, 0xd4, 0xf1, 0x5f, 0x88, 0x6, 0xc1, 0x63, 0xbb, 0x3a, 0xc8, 0xd4, 0x8f, 0xe9, 0xdd, 0xbb
CRC valid!

netid1 is 16, netid2 is 2008, check skipped
rx msg: 0xff, 0xff, 0xff, 0xff, 0x9, 0x60, 0x54, 0x5e, 0xfa, 0x35, 0x6e, 0x2c, 0xe0, 0x8, 0x0, 0x28, 0x74, 0x0, 0xa2, 0xb0, 0x2d, 0xac, 0xba, 0xef, 0x72, 0x37, 0x8a, 0xa8, 0xca, 0x42, 0x80, 0x25, 0xa2, 0xc3, 0xf3, 0xc4, 0xdc, 0xab, 0x5a, 0x77, 0x47, 0xab, 0x4, 0x7, 0x67, 0xbc, 0x28, 0x23, 0x9f, 0xd4, 0x3b, 0x42, 0x44, 0xa8, 0x2c, 0x28
CRC valid!

netid1 is 16, netid2 is 2008, check skipped
rx msg: 0xff, 0xff, 0xff, 0xff, 0xdc, 0x80, 0xa4, 0x88, 0x88, 0xf2, 0x3e, 0x65, 0xe0, 0x8, 0x0, 0x22, 0x1d, 0x13, 0x4a, 0x17, 0x4b, 0x22, 0x59, 0x3, 0xc9, 0x35, 0x5b, 0xe, 0x94, 0x4b, 0xd3, 0x28, 0xfd, 0x79, 0xa3, 0x41, 0x9c, 0x6, 0x8, 0x8b, 0x3c, 0x68, 0x2d, 0xad, 0x65, 0x35, 0x9f, 0x4e, 0x83, 0xcf, 0x28, 0x36, 0xfa, 0xef, 0x5a, 0xd8, 0x53, 0xd3, 0xac
CRC valid!

netid1 is 16, netid2 is 2008, check skipped
rx msg: 0xff, 0xff, 0xff, 0xff, 0xcf, 0xd7, 0x4e, 0xb, 0xf0, 0x1b, 0x76, 0x79, 0xa1, 0x13, 0x0, 0x68, 0x7e, 0xcc, 0x8c, 0xf0, 0xa0, 0x4d, 0x50, 0x69, 0x3a, 0x14, 0xa3, 0xfb, 0x29, 0xd3, 0xbe, 0x6e, 0x4b, 0x1e, 0xdf, 0xf5, 0x8a, 0x85, 0x64
CRC valid!
```

radio 101

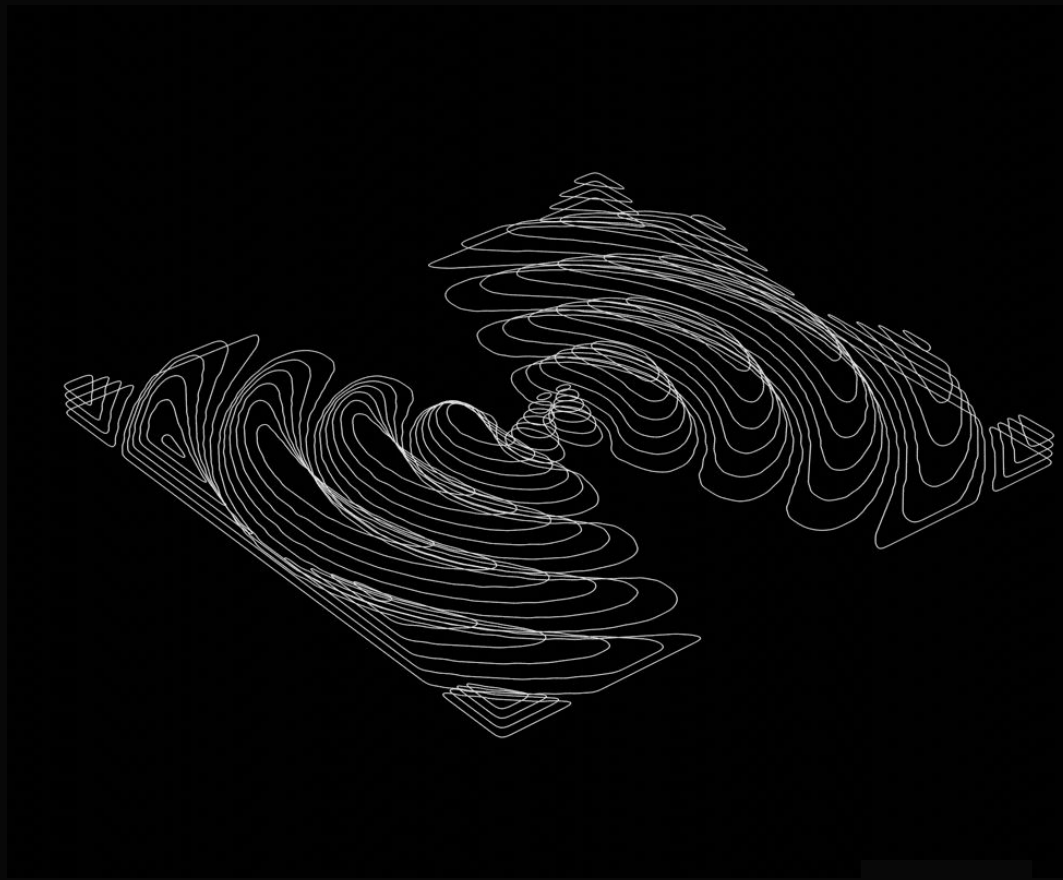
a radio **transmitter (TX)** is a photon gun, constantly emitting electromagnetic radiation at a desired **frequency**.

the speed of the wave equals the frequency of the wave times the **wavelength**

1 hertz (Hz) = 1 cycle/second
= 1 wave/second
1 megahertz (MHz) = 1,000,000 Hz

like all electromagnetic radiation, radio waves travel at the speed of light (300,000,000 m/s in vacuum)

(Wavelength in meters) * (Frequency in MHz) = 300
divide 300 by one to get the other



Animation of radio waves from dipole antenna rippling through 3D space (https://blender.stackexchange.com/questions/304616/creating-an-animation-of-radio-waves)

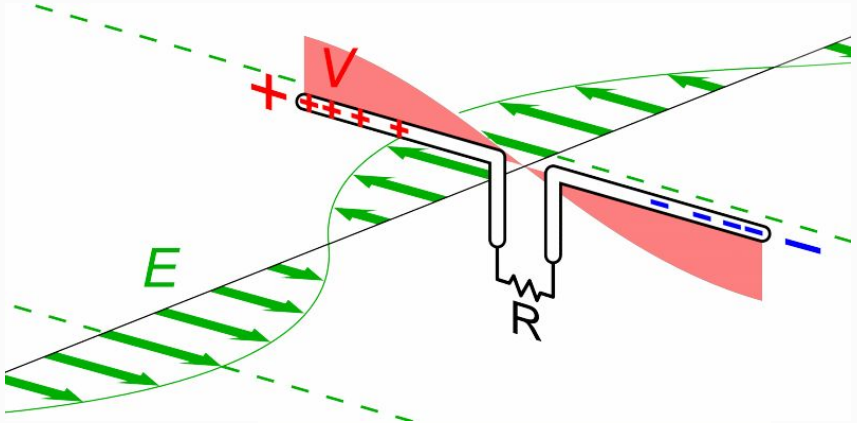
radio 101

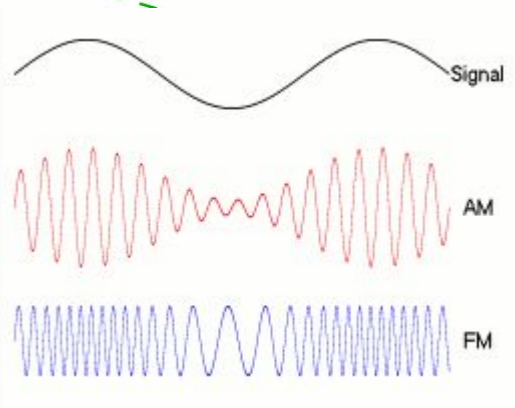
a radio antenna **tuned** to the right frequency can receive the electromagnetic wave and convert it into **electric current**

a radio **receiver (RX)** is a circuit that decodes the information encoded in the received wave, and processes/outputs that information

some ways to encode information in a **carrier wave:**

- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- LoRa (too complex to draw easily, but it's just another way to encode information!)





Top - https://upload.wikimedia.org/wikipedia/commons/e/ea/Dipole_receiving_antenna_animation_6_300ms.gif
Bottom - https://en.wikipedia.org/wiki/Frequency_modulation

Why Meshtastic?

[vs Signal] resilient and decentralized

[vs ham radio] relatively cheap, encrypted and no license required

cheap radio experimentation platform whose potential is maybe 1% explored

novel comms technique that's SUPER easy to hack!!

Why Not Meshtastic?

known limitations of **Meshtastic security model (https://meshtastic.org/docs/about/overview/encryption/limitations/)**:

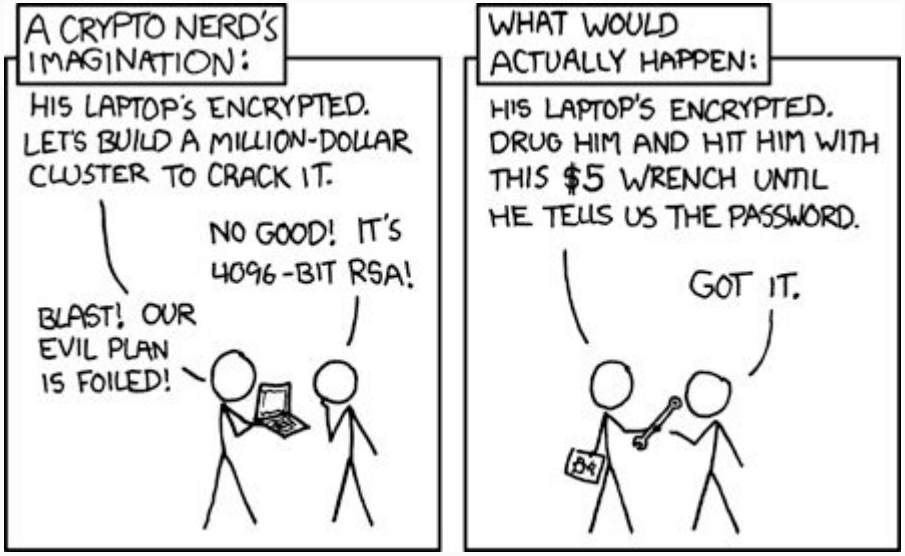
- No authentication
- Unattended nodes are insecure
- No perfect-forward secrecy

reliability concerns
current limitations of client software

Meshtastic is great for building a new 915MHz mesh network, Meshcore is better for efficiently using an established network

TL;DR - *we are building comms that work when all else fails*

Use Signal for E2EE
Use established comms for critical matters
Learn how Meshtastic/Meshcore/LoRa works **before** all else fails!



DEMO –
Spoofing Node Hardware ID

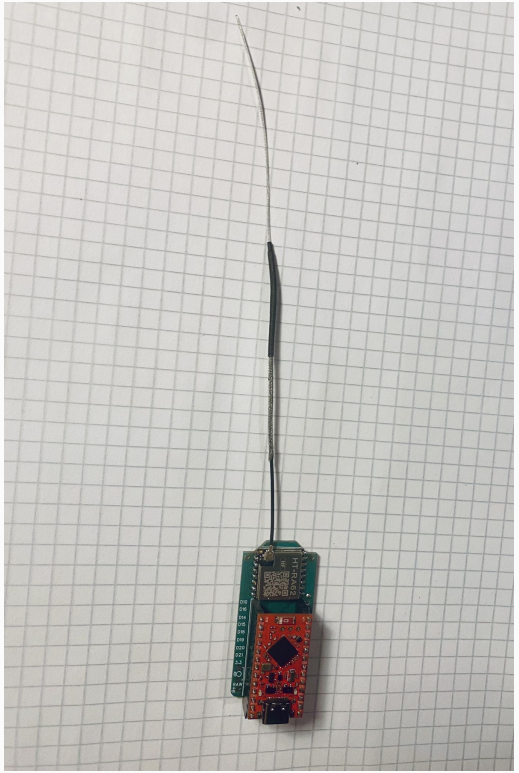
Building Meshtastic Nodes for Cheap

Hampton Bay Solar Node Hack
- Use a $10 Lowe's Solar Light for its:
  - 1W Solar Panel
  - 2+ Ah battery
  - Battery Management System
  - Waterproof case

FakeTec for DIY Bluetooth/LoRa
- Replace an $18 Heltec T114 with $9 DIY
  - $2 SuperMicro NRF52840 Bluetooth "Redboard"
  - $4 HT-RA62 915MHz LoRa Radio
  - $2 green board to JLCPCB
  - <$1 Various 1206 surface mount components

Battery packs with 9x capacity/$ ratio
- Solder JST cables to nickel strip, spot weld nickel strip to 18650
- Shrink wrap tubing to hold the cable in place
- LiPo thermal runaway is hellfire, use LiFePO4 if possible

Stevens Stinger v2 - Modify $1 1.13mm coaxial cable into <1.5 SWR



DEMO -
Controlling Nodes Programmatically via Meshtastic SDK

Monitoring Meshtastic Networks

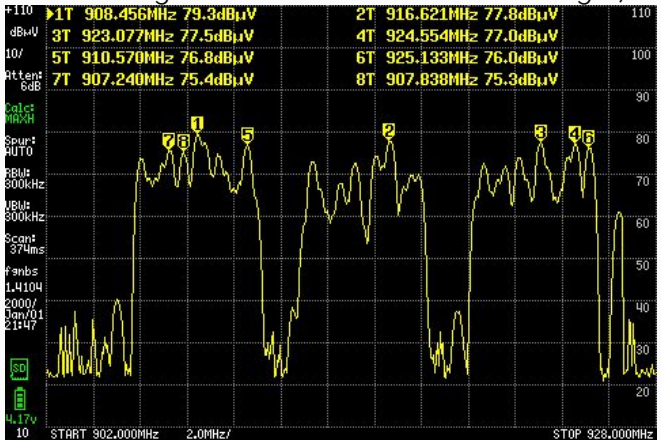
RF measurements with TinySA (foxhunting, spurious emissions testing) and NanoVNA (antenna quality testing)

These tools make it straightforward to build very good antennas at home!!

Give intuitive understanding of how radio waves propagate and how certain RF attacks/techniques function (triangulation)

Node admin via publickey

Shoutout — Catgirl Works Matrix Bridge, https://awoo.corg.ski

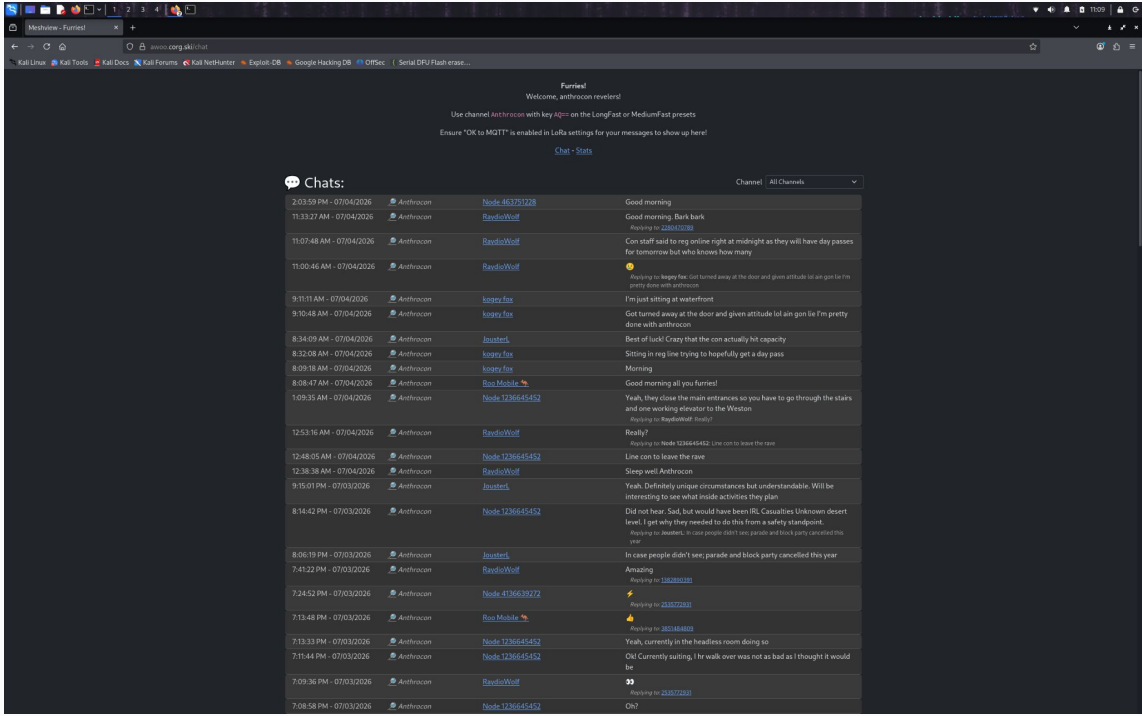




Monitoring Meshtastic Networks

Can forward messages via MQTT Bridge frequency presets (“Anthrocon” channel, also shown at right on https://awoo.corg.ski) Bridge platforms (Catgirl Works Matrix Bridge bridges Meshtastic+Meshcore)

Remote node admin via node publickey



Conclusion

Meshtastic is ultra-resilient decentralized encrypted comms currently exploding in popularity

The project/technology/effort needs experts of all kinds:
- Assembling + programming nodes
- Researching new node designs
- Hosting onboarding workshops
- Coordinating between projects
- Building client software
- Security testing
- Whatever you're good at, we probably need it!

Lots of great projects to join, especially here in Pittsburgh!
- WPAmesh
- Protomesh Collective
- Hacker's Guild

Thank You! Questions?

Contact me at
dani@societyofhackers.com